Chapter 54 - Electro-Magnetic Spectrum
Calculus Physics II

The Electromagnetic Wave Spectrum The EM spectrum is the complete range of all types of electromagnetic radiation, organized by their frequency, wavelength, or energy. All forms of EM radiation travel at the speed of light in a vacuum but differ in how they interact with matter based on their wavelength and frequency. The frequency of the EM wave's ranges from a few Hertz for radio waves to 10^{24} Hz for high energy gamma rays. The types of EM waves in the spectrum listed below are from lowest energy to highest energy.

• Radio waves • Microwaves • Infrared waves • Visible light • Ultraviolet light • X-rays • Gamma rays

Radio Waves

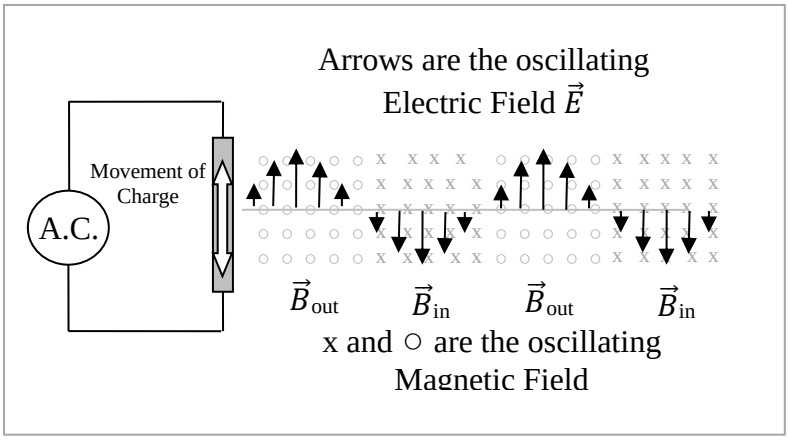
Radio waves – are the lowest frequency and longest wavelength of waves in the electromagnetic spectrum, they are below the microwave spectrum. James Maxwell predicted the existence of Radio waves and Heinrich Hertz is the one that actually verified their existence using a spark gap to generate the waves and a second gap to detect them. Radio waves are not visible to the human eye.

Range of radio waves Frequency range: up to 300 GHz
Wavelength range: up to 1 cm

Radio Waves are created by: movement (acceleration) of charge, most commonly by the acceleration of electrons.

How are radio waves created and what are there sources – Radio waves are created when electrons are accelerated. When a charged particle is accelerated it creates a changing magnetic field, both the electric and magnetic field propagate out from the accelerating charge as a wave. Oscillating charge creates changing electric field, moving charge creates changing magnetic field. These fields radiate out from the antenna.

1) When the electric field generated by the ac source is pointed up the electrons are flowing downward in the opposite direct as the $\vec{E}$ field, this downward current produces a magnetic field out of the page (#2 right hand rule).

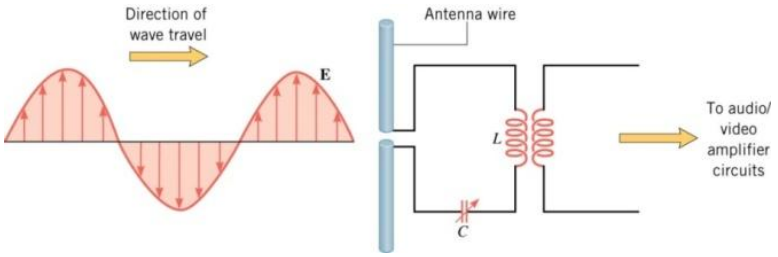


2) When the electric field changes and points downward this drives the electrons upward which flips the magnetic field direction into the page
3) This process keeps going as long as the ac power source is oscillating the charge.

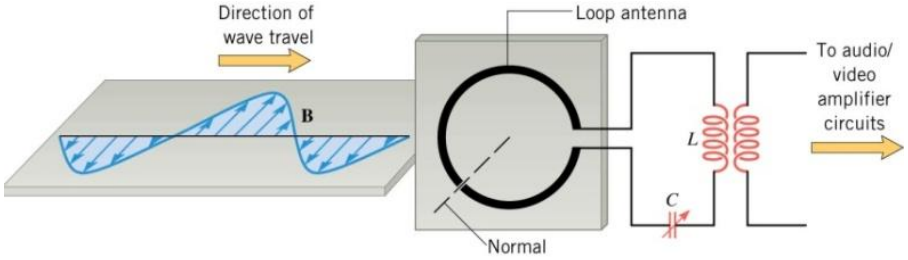
How do we detect radio waves?

Antenna – is an electrical device (conductor) that converts electrical signals into EM waves (emitter) or converts EM waves into electrical signals (detector).

Receiving EM Waves via Electric Field – a charge in an electric field will experience a force and if that charge is free to move, a positive charge will moves in the direction of the electric field. Because an electrical conductor (metal antenna) has charge that is free to move, the electric field from the electromagnetic wave will cause the charge to move with the electric field. The charge will move back and forth with the same frequency as the electric field in the electromagnetic wave, if there is a system of circuits to detect and amplify the EM wave then information can be transferred from the EM wave to the device. This is a simple radio receiver or cell phone. The electric field in the EM wave is tracked exactly by the moving charge in the antenna.



Receiving via Magnetic field – instead of using the electric field of the EM wave, the magnetic field can be detected by a wire loop (antenna), a changing magnetic field inside a wire loop creates an induced current in the loop at the same frequency as the EM wave. An electronic device can detect and amplify the signal. This is used in detecting AM radio waves.



Applications of Radio waves – most applications of radio waves are for communication and Radar.

Information carried by FM, AM and digital EM waves - A electronic signal is transmitted by way of Electromagnetic waves (radio waves) to a receiver (antenna, receiver and amplifier) that converts this electronic signal back into a wave.

Modulation – is the act of changing the properties of a wave (carrier wave) to transmit information.

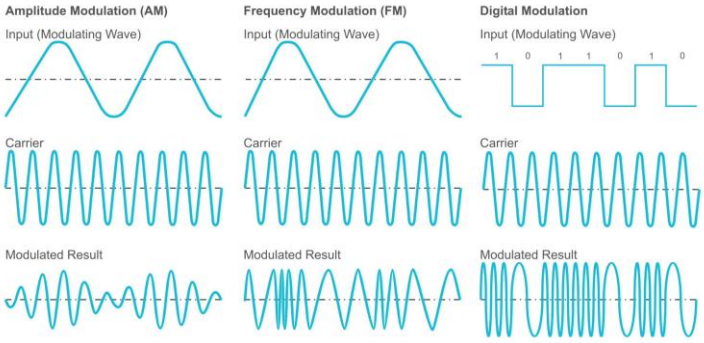
Carrier Wave – is the wave form that is modulated (changed) and caries the information.

The electronic information can be transmitted by three methods.

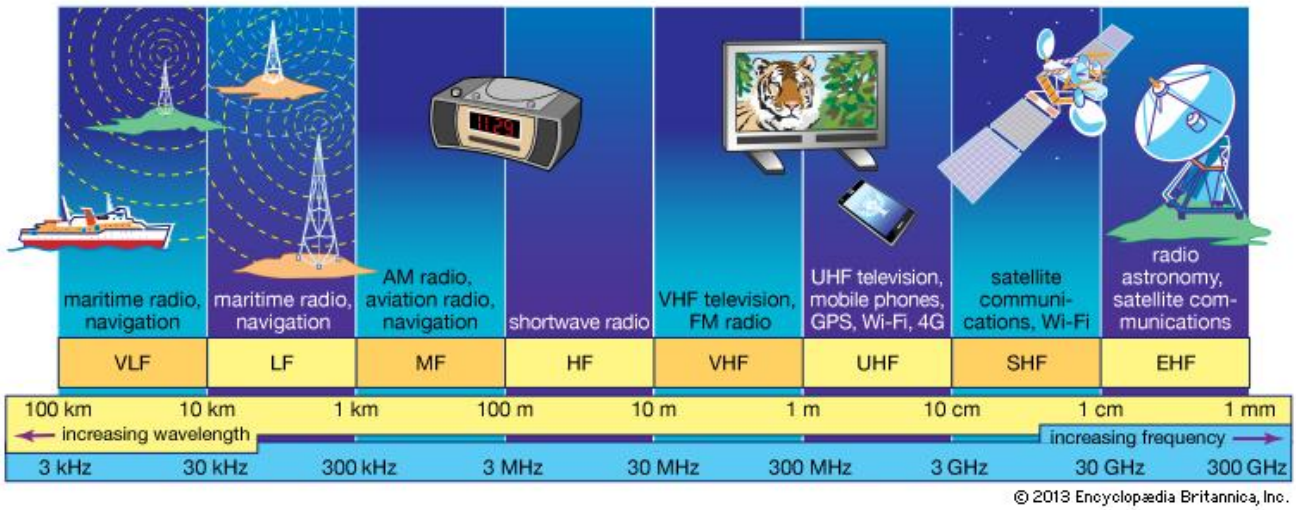
AM – Amplitude modulation (AM) transmits the information by changing the amplitude of the carrier wave.

FM – Frequency modulation (FM) transmits the information by changing the frequency of the carrier wave.

Digital – converts the signal into zero's and one's and sends a pulse for one and no pulse for zero using the carrier waves amplitude.

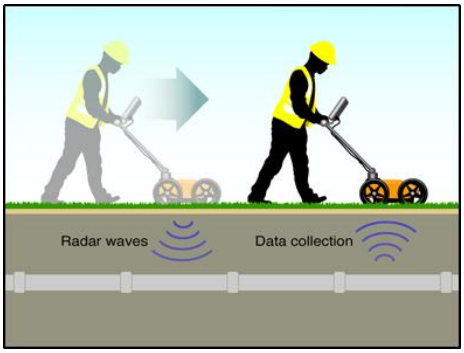


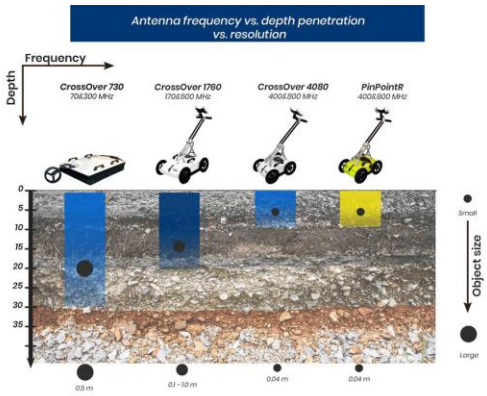
The bands in the radio wave region of the spectrum are controlled by the FCC and have set applications for set frequency ranges.



FCC – Federal Communication Commission – An agency of the United States Government that regulates the transmission of EM waves for radio, TV, wireless, wired, satellite and cable.

Ground Penetrating Radar (GPR) - Pulses of radio waves are sent into the ground, as the waves move though the earth they are reflected by abrupted changes in the density of the ground (pipes, tunnels, wires, ground layers, etc. …). A sensor on the GPR detects the amplitude and time of the reflected waves return, from this the system can determine depth and densities of objects and is converted into images of the object under the ground. The GPR uses radio waves from 10 MHz up to 2 GHz. The different radio frequencies affect the penetration depth and the resolution of the images.





Microwave Radiation

Microwaves are electromagnetic waves that have a higher frequency than radio waves but lower frequency than infrared radiation. Microwaves are the highest energy waves that can be produced by oscillating electric current.

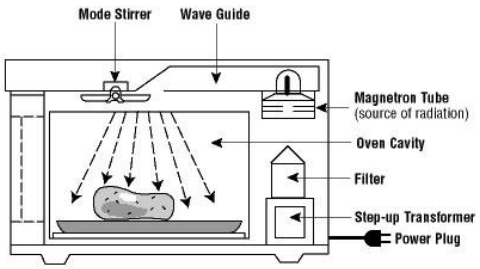
- Microwaves are not visible to the human eye.

Range of microwaves: Frequency range: 10^8 Hz to 10^{12} Hz
Wavelength range: 1 cm to 25 μm

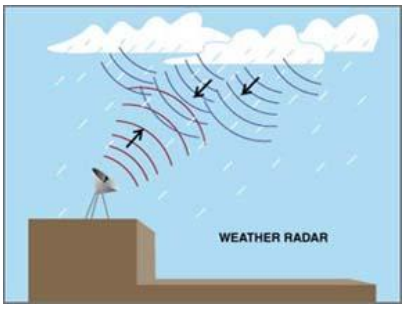
Microwaves are created by: the acceleration of electrons similar to radio waves except at a much higher frequency.

Applications of Microwaves

Microwave ovens – use a Magnetron to generate microwaves radiation that is guided to the oven by rectangular metal wave guides. Once the microwaves enter the oven, they go through a reflecting fan that moves the microwaves around the oven so that there are minimal "hot spots" in the cooking. The Microwaves are absorbed by water and this energy is converted to thermal energy (heat). Microwave ovens operate at 2.45 Ghz. They do not use nuclear energy but EM waves!



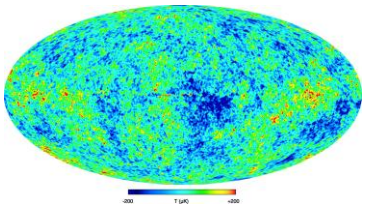
Radar – **Ra**dio wave **d**etection **a**nd **r**anging – both radio waves and microwaves are used in Radar depending on the application. A signal is sent out from the source, and it reflects off of objects such as clouds, planes and the reflected signal is detected. The time it takes to return determines the distance the object is away and using the Doppler Effect (change in frequency due to moving object) the velocity of the object can be determined also.





Communication – Wireless communication – most wireless communication (cell phones, cell towers, Wi-Fi, cordless phones, Bluetooth, etc. …) uses microwaves to send information. Similar to radio waves the FCC controls the range of frequency that can be used for certain applications so that there not inferencing with each other and their applications.

Cosmic Mircowave Background (CMB)- is the radiation left over after the Big Bang when the universe was formed. Theory pedicts that at the beginning of the formation of the universe all matter was in a very hot plasma state but as it cooled protons and electrons started to form releasing large amounts of high energy radiation which over time (and space) expanded into the microwave region of the EM specturm. which are dectected today. CMB radiation is thoroughout all the universe and is not uniform but maps the initial expansion of the universe.



False color map of the Cosmic Microwave Background (CMB) of the universe.

Infrared Radiation

Infrared radiation (IR) – is between microwaves (lower frequency and visible light (higher frequency) and is associated with the vibrational emission due to thermal energy of materials.

- Infrared light is not visible to the human eye

Range of Infrared radiation Frequency range: 10^{12} Hz to 10^{14} Hz
Wavelength range: 25 μm to 750 nm

Infrared radiation is created by: vibrations in the charge of the atoms and molecules due to thermal energy. Any object that has thermal energy will emit infrared radiation. The higher the temperature of an object the more infrared EM waves it will emit.

Applications of infrared radiation

Remote for tv – remote controls for TV's use infrared LED's to communicate with the box, infrared has similar properties to visible light, this is why there must be a "line of sight" to get the remote to work, IR is blocked by object.

Heating and cooking food – infrared radiation is thermal radiation therefore it can be used to keep food warm and cook food; heating lamps use infrared radiation.





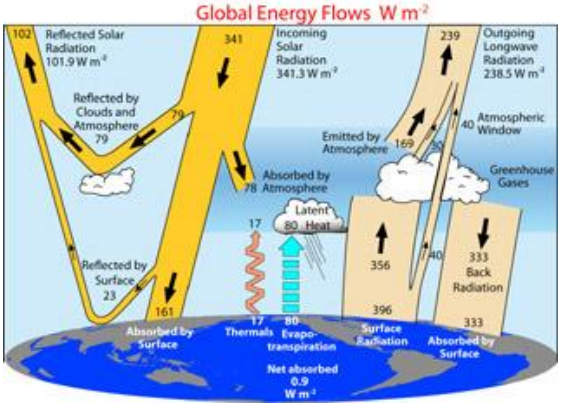


Thermal imaging and sensing – because IR is thermal radiation the temperature of objects can be sensed using this radiation. A touchless thermometer uses the infrared emitted by the forehead to measure the temperature of the body.



The "Green House" effect – for a real green house with glass windows, when the EM spectrum from the sun goes through glass is absorbed by the objects inside the house this heats up the objects and the objects emit infrared radiation. Most of the infrared radiation is reflected by the glass back into the house therefore the house heats up due to more energy coming in the house than leaving.

A similar effect is happening on earth, more energy from the sun is getting through the atmosphere than is radiating back out because of "greenhouse" gasses that trap the energy near the surface. The more greenhouse gasses that are present the more energy is trapped and the hotter the earth will get.



Transparency in the infrared - The first image is a person holding a trash bag shown in visible light, the second image is the same person and bag shown in the infrared wavelengths, the trash bag is transparent in the infrared.

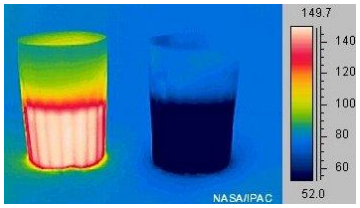


<u>Using Infrared Radiation</u> - Picture taken with an Infrared Camera, what looks white is warm and what is dark is cooler, the camera "see" the temperature of object. The camera uses false color to show temperature differences. (color is a visible light property so infrared does not have "color")

- When an object has a temperature higher than its surroundings the object emits a net radiant power.

The red and blue cup in the top image is have hot and cold liquid in them but in the visible spectrum you can not tell by looking but in the infrared image below the red cup appears hot (white) and the blue cup appears cold (dark blue).





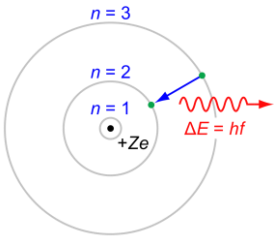
A house imaged in the infrared shows temperature differences on the outside, the red color is warmer than the green and the blue is cold. This image shows where thermal energy is escaping from the house.

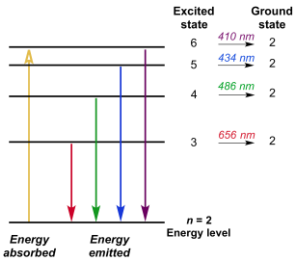




An infrared image of people walking up to a car shows that the people are warmer than their environment and the car has been running, the hood is warm and the reflection of infrared radiation can be seen under the car.

EM waves from Atoms/molecules - Electromagnetic waves can also be created by the electrons in the atom moving to a lower energy state in the orbitals, the energy from the change in orbitals is given off as EM waves. The larger the change in the energy states the higher the EM wave's energy producing higher frequency EM waves. The atom produces EM waves from visible, ultraviolet to x-rays.





- ❖ Molecules give up energy in a similar way as atoms, but the energy change is less therefore the wavelength is longer (infrared region).
- ❖ Visible light is produced by the atomic transitions of the electrons in the atom therefore they have more energy than infrared radiation.

Visible Spectrum

Visible light – is part of the electromagnetic spectrum and is only special because human eyes can detect this range of EM radiation.

- We see this region of the EM spectrum because human eyes are sensitive to visible light! If we see it, there must be visible light coming from it.

Range of visible light: Frequency range: 7.5×10^{14} Hz to 3.8×10^{14} Hz
Wavelength range: 400 nm to 750 nm

Visible radiation is created by: transitions of elections in the atom or by objects with large thermal energies.

Visible Spectrum - In the visible spectrum red light has the lower frequency so it has the lower energy, blue light has the higher frequency, so it has the larger energy.

R	Red	Lower Energy	610 to 750 nm
O	Orange		590 to 610 nm
Y	Yellow		570 to 590 nm
G	Green		500 to 570 nm
B	Blue		450 to 500 nm
I	Indigo		425 to 450 nm
V	Violet	Higher Energy	400 to 425 nm

- ✓ Increasing frequency increasing energy.
- ✓ Decreasing wavelength decreasing energy.

Photopic – the part of EM spectrum that humans can see with their eyes.

Applications – there are many applications for visible light including imaging, communications, health and healing, etc. …

Ultraviolet Spectrum

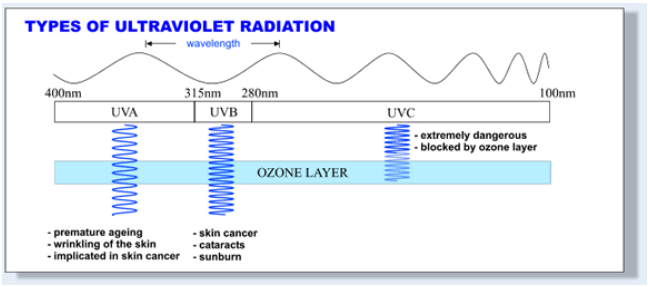
Ultraviolet Light – is the part of the electromagnetic spectrum that is just above the visible spectrum and has more energy associated with it than visible light.

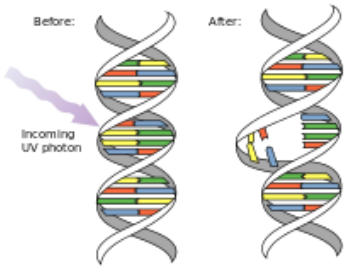
Range of Ultraviolet Light: Frequency range: 7.5×10^{14} Hz to 3×10^{16} Hz
Wavelength range: 400 nm to 1 nm

Ultraviolet radiation is created by: high energy electron transitions in the atom or object with very large thermal energies.

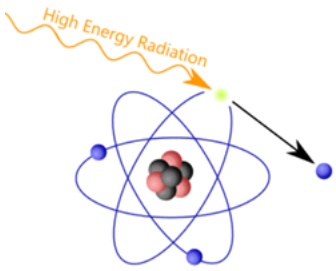
Ultraviolet radiation is divided into three bands UVA, UVB and UVC

- UVC has the highest energy and is the most dangerous but does not penetrate our atmosphere, it is absorbed by the Ozone Layer. UVC is Ionizing radiation.
- UVA and UVB is what gives our skin a "tan" or sunburn but can also cause cancer. Is not Ionizing radiation but can break molecular bonds.
- UVB is the EM waves from sun that create vitamin D in human bodies.
- UV is also used for sterilization (kill germs) in air and water.

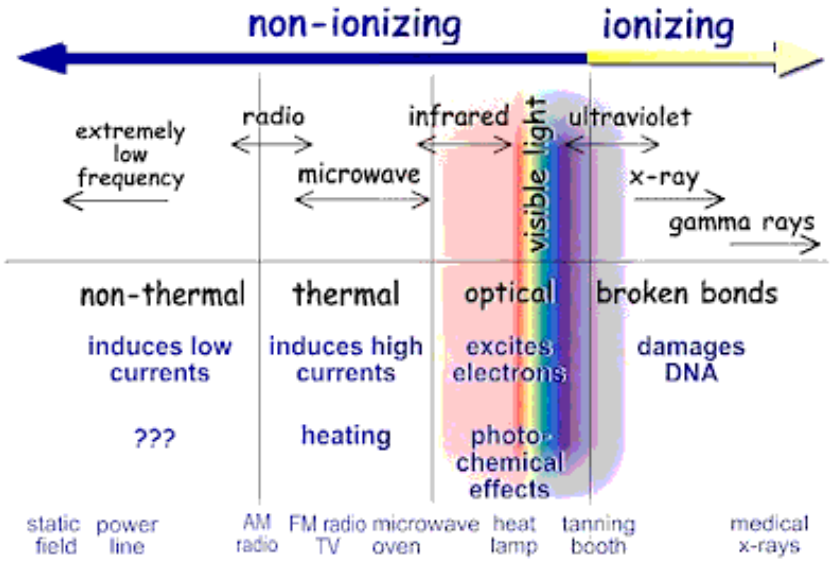




Ionization Radiation - if the electromagnetic wave has enough energy, this radiation can separate electrons from their atoms in material creating ions or break the bond of atoms in molecules which make it dangerous for living tissue. UVC, X-ray and Gamma rays are all ionizing radiation.



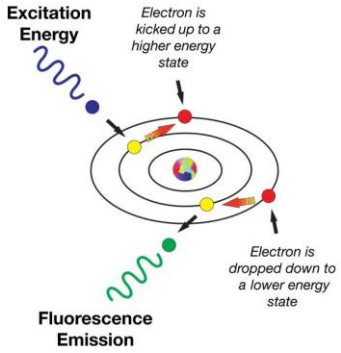
Ultraviolet light (UVC) is Ionization Radiation - because of the high energy of the UV light, this radiation can separate electrons from their atoms creating ions. This effect has its applications and can be useful but for human skin it can cause damage to cells.



Curing applications - UV radiation can be used for an initiator and curing polymers for applications in dentistry and industry.

Fluorescents - is the emission of visible light due the absorption radiation from other parts of the EM spectrum. Because of UV radiations high energy, it will cause some objects to fluoresce in the visible spectrum. Backlights are UV lights that make white paper or white cotton fluoresce. There are also fluorescent dyes that are used in medical and industrial applications.





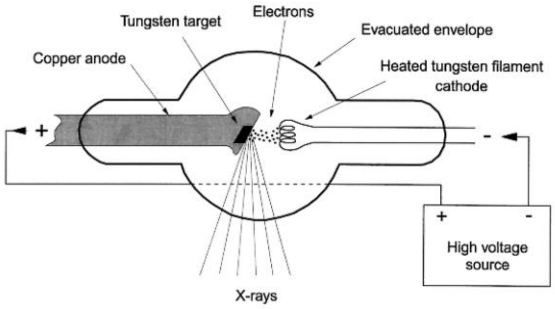
X-ray Radiation

X-ray radiation – is a very high frequency radiation and has the second highest energy of the EM spectrum (expect for gamma rays). X-rays will go through most nonconductive solid materials. The "X" in x-ray indicated that scientist did not know what they were. X-rays are ionizing radiation therefore they have adverse effects on tissue but because they penetrate most materials, they are used to image bone and other hard substances in the human body.

Range of X-ray radiation: Frequency range: 30×10^{18} Hz to 30×10^{15} Hz
Wavelength range: 1 nm to 1 pm

X-ray radiation is created by: very high energy electron transitions in the atom or extremely large thermal energies for example massive stars and super nova's emit x-ray at temperature of a million Kevin and above. They can also be created by Synchrotrons, very fast charged particles (electrons or protons) moving in a circle by way of a magnetic field.

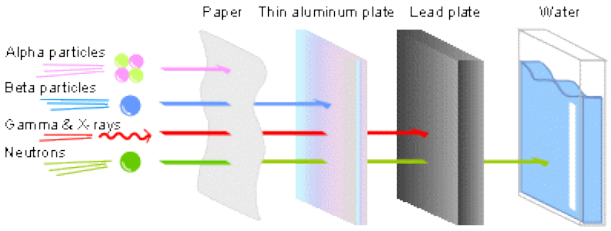
Generation of x-rays – (**Cathode ray tube CRT**) the most common method of generating x-rays is to use a high voltage source (large Electric fields) to eject high energy electrons that hit a target, the energy of the electrons is transferred to the target and the target emits x-rays.



Soft x-rays – wavelength of 0.2 to 10 nm lower energy and closer to ultraviolet radiation.

Hard x-rays - wavelength of 0.2 to 0.01 nm, they are higher energy x-rays that are closer in energy to gamma radiation. This is the x-ray radiation used in medical applications.

Transmission of x-rays trough materials - X-ray and gamma rays will penetrate through most solid non-conducting materials and can penetrate conducing material if it is thin enough.

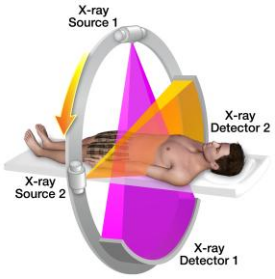


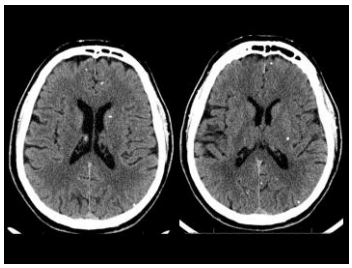
The amount of x-ray radiation that is able to penetrate material depends on the density of that material therefore detecting the number of x-rays that get through an object will give an image of the density of that object, which is shown in the x-ray of the dog and hand below.





CT Scan – Computer Aided Tomography scan – small, focused beams of x-rays are scanned over the entire body or area of interest, x-ray sensors detect the transmitted radiation, and the computer creates a 3-D and cross-sectional image from the scans. The scans are very sensitive and can detect very small changes in density such as fluids, cartilage, or bones.





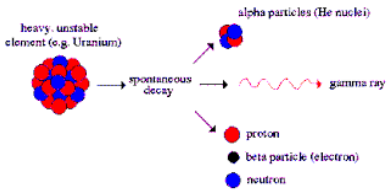
Gamma Rays

Gamma rays – when an atomic nucleus decays it emits three types of radiation, alpha, beta and gamma. (Alpha and Beta radiation are NOT EM waves). Gamma radiation is the most penetrating and is the highest energy EM waves. The range of Gamma rays has overlap with x-rays, but the difference is the source of the radiation, gamma rays come form the nucleus of the atom.

Range of Gamma Rays: Frequency range: 10×10^{18} Hz and up
Wavelength range: 1 pm and smaller

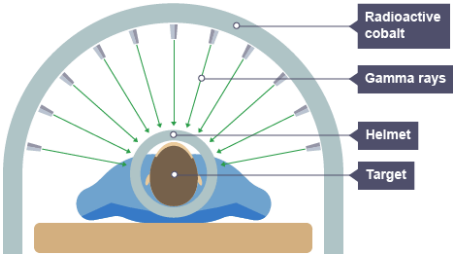
Gamma rays are created by: nuclear energy transitions, black holes, super nova's and matter/antimatter annihilation.

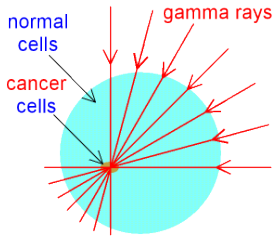
Atomic Nucleus and EM Waves – The nucleus of the atom can emit electromagnetic waves during radioactive decay. Part of the change in energy of the nucleus is given off as high energy EM waves (gamma rays)



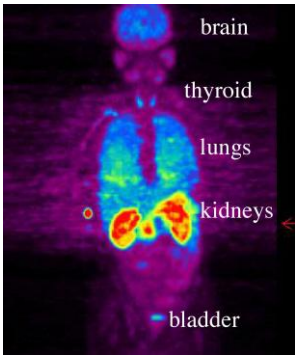


Cancer treatment (Radiotherapy) – gamma rays can be used for the detection and treatment of cancer because gamma rays are ionizing radiation they can be used to kill cancer cells and can be detected in the blood stream as a tracer element.

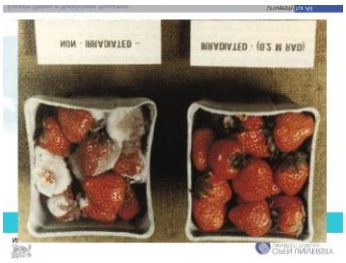




PET scan – Positron Emission Tomography – A radioactive isotope called a tracer is injected into the body, The body absorbs the isotope, when the isotope decays (Beta Decay) it produces an anti-mater particle called a positron (anti-matter of the electron), this particle is very short lived and annihilates with nearby electrons producing a gamma ray. The gamma rays are detected by sensors and a computer is used to create images from the gamma radiation. The false color indicates where the isotopes where absorbed the most and the least.



Sterilization - gamma rays can be used to sterilize food and equipment.



Sources of gamma rays in our universe are from exploding stars, neutron stars, black holes

Lightning creates gamma rays due to the high energy of the discharge of the current (moving charge) through the atmosphere.

EM spectrum transmission through Earth's Atmosphere – not all of the EM spectra will make it through the earth's atmosphere, the higher energy part of the spectrum does not (UV, x-ray or gamma rays). Visible light and near infrared light makes it through the atmosphere but most of the infrared does not. There is a band in the radio wave spectrum that the atmosphere is transparent to, and it is used for communication with satellites and for looking at the universe using radio telescopes.

This is one reason that some telescopes are placed in orbit around the earth and outside of the earth's atmosphere, it so the satellites can "see" the universe using the entire EM spectrum.

